

Matrix functions and stability

Uniform polynomial bounds for products at joint spectral radius one

Difficulty: challenging **Importance:** interesting to the community

Status: open; Conjecture L3 in the source below

Context and notation

Let $\mathcal{H}_d$ denote the nonempty compact subsets of $\mathbb{C}^{d \times d}$.

The joint spectral radius of a nonempty compact set $\mathcal{M} \subset \mathbb{C}^{d \times d}$ is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

Problem statement

Does each $d \geq 1$ admit $\Theta_d > 0$ such that every $\mathcal{M} \in \mathcal{H}_d$ with $\hat{\rho}(\mathcal{M}) = 1$ satisfies

$$\|A_k \cdots A_1\|_2 \leq \Theta_d (Lk)^{d-1}, \quad L = \max_{A \in \mathcal{M}} \|A\|_2,$$

for all $k \geq 1$ and all $A_1, \dots, A_k \in \mathcal{M}$?

Reference and status evidence

Epperlein and Wirth, [The joint spectral radius is pointwise Hölder continuous](#), §2, Conjecture 3 (L3). Lemma 27 proves dimension two. The constant above must be independent of the family.

Common follow-up screen for MF-05–MF-07

Searches combining “joint spectral radius” with “local Hölder”, “Lipschitz lower”, “trajectory bounds”, the authors’ names, and 2025/2026 found no later resolution. These searches supplement the explicit 2025 conjectures; they do not establish exhaustiveness. The three entries are separately named assertions in the source, not a count of dimensional cases.